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Penalized GANs with latent perturbation for robust shilling attack generation in recommender systems

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Abstract
Shilling attacks pose a significant threat to the integrity and reliability of recommender systems by injecting fake user profiles to promote or demote targeted items. Existing generative approaches often suffer from unstable training dynamics and limited realism in the synthesized profiles. In this paper, we propose PGAN, a novel Penalized Generative Adversarial Network enhanced with latent space perturbations to generate high-quality, diverse, and undetectable shilling attack profiles. PGAN incorporates a gradient penalty to stabilize discriminator training and applies controlled noise perturbations in the generator's latent space to improve robustness and attack diversity. We evaluate PGAN on real-world datasets and demonstrate that it consistently outperforms traditional statistical attacks and baseline GAN-based models across multiple evaluation metrics, including Hit Ratio@K, Prediction Shift, and attack success rate. Experimental results also confirm the realism of the generated profiles through similarity analysis with genuine users. Our proposed model outperforms traditional and state-of-the-art methods, achieving HR@10 scores of 0.2051 and 0.2076 on the MovieLens and Amazon datasets, respectively.

Keywords Shilling attacks, Recommender systems, GCN, GAN, Perturbations, Gradient penalty

1 Introduction

Recommender systems are widely adopted in various domains and online services, including online shopping and book recommendations, providing effective solutions to the overwhelming amount of online information. They aim to enhance users' experiences by generating personalized recommendations based on their preferences []. Additionally, they benefit businesses by increasing sales and fostering customer loyalty. Customers tend to return to platforms that effectively meet their needs. For example, collaborative filtering recommender (CF) [2] is one of the most widely adopted techniques for online products. It identifies users with similar preferences and generates recommendations based on similarity metrics. The widespread use and openness of recommender systems have led to an increasing number of attempts to manipulate them for malicious purposes. Despite the effectiveness of recommender systems in many fields, they are vulnerable to shilling attacks, also known as poisoning or profile injection attacks [3]. These attacks are executed by injecting fake user profiles with altered ratings, misleading the recommender systems into generating recommendations in a skewed manner [4, 5]. That manipulation is aimed at either promoting or demoting a product, for instance, promoting the attacker's products or demoting those of their competitors. Shilling attacks are generally categorized into low-knowledge and high-knowledge attacks. Highly knowledgeable attackers possess detailed knowledge of the incorporated system and the user-item interaction matrix. At the same time, low-knowledge attacks can be executed without detailed knowledge regarding the systems, yet they can still cause significant damage with less information [6]. For example, Fig. 1 illustrates two scenarios in an item-based collaborative filtering recommender: (a) and (b). In (a), the recommender system performs its genuine function by generating relevant and similar items based on item-item similarity. In contrast, in (b), the attacker introduces biased profiles and interactions to target specific items, resulting in users receiving irrelevant recommendations that undermine the system's integrity. Several studies in the literature focused on identifying and detecting shilling attacks through various algorithms, while others have worked on developing robust recommenders that are more resilient

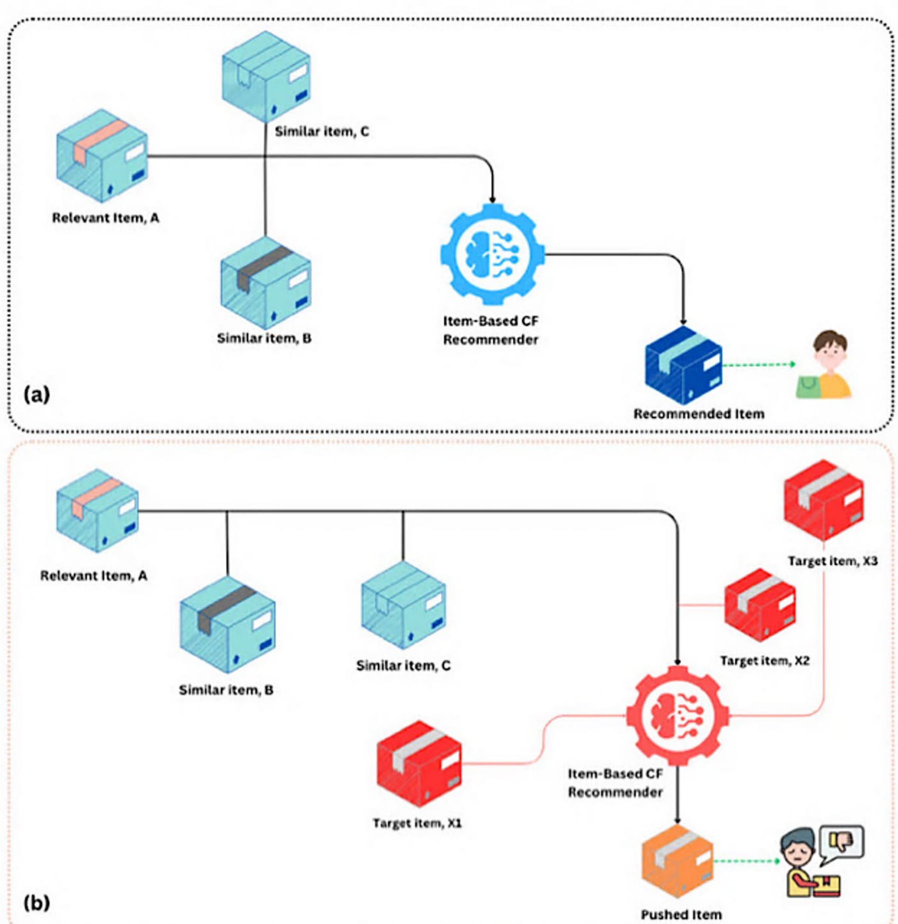

Fig. 1 Item-based CF (**a**) Normal flow, (**b**) Shilling attack

to manipulation [7, 8]. Shilling attacks were generated using simple techniques, making them easy to deploy and detect.

More advanced methods, now using sophisticated models, are being adopted, which generate more effective attacks; however, these models can be complex and have a high cost. It is challenging to balance the effectiveness and the cost of generating shilling attacks, especially since most attackers rely on prior knowledge of the used recommender system and user-item data [9]. Most attack models focus on using simple heuristics, such as assigning high or low ratings to target items while rating other items (filler items) randomly around the system's average rating [10]. A successful attack requires a large number of injected fake profiles, which can be very computationally expensive to apply. Generative Adversarial Networks (GANs) can be used to generate powerful shilling attacks by training diverse attack profiles that deceive real users [11],. However, the adversarial nature of GAN training often leads to instability, such as convergence issues or mode collapse, hindering the generation of effective attacks. In this paper, we introduce a novel attack approach that leverages the generative capabilities of GANs. The PGAN model with noise perturbations is proposed for generating realistic user profiles in recommender systems, simulating undistinguishable shilling attacks compared to heuristic attack models. In the PGANs, Graph Convolutional Networks (GCNs) are used to model user-item interactions by transforming the data into a bipartite graph, aiming for improved representation. The resulting output of the GCN is then fed into the PGANs pipeline, with the addition of noise perturbation techniques for more realistic-looking fake profile generation. The model is tested using two datasets: MovieLens 100 K [12] and Amazon Electronics [13], where it is evaluated against traditional shilling attacks and state-of-the-art models. Our main contribution can be summarized as:

- We propose PGAN, a novel GAN-based model enhanced with penalization and latent perturbation for generating realistic and robust shilling attacks.
- We merged the strengths of GCN, combined with a bipartite graph, into a penalized GAN framework to model user-item interactions.
- We introduce a gradient penalty in the discriminator to stabilize the adversarial training process.
- We apply controlled Gaussian perturbations to the generator's latent input space to improve profile diversity and reduce detectability.
- We compare PGAN against baseline statistical and generative shilling attacks, demonstrating its superiority in promoting target items.

The remainder of the paper is organized as follows: Section II discusses the background of shilling attacks, including the commonly used standard attack types; Section III highlights the existing work in the literature; and Section IV explains the proposed methodology. Section V shows the experimental results obtained. Sections VI and VII represent discussion and conclusion, respectively.

2 Background on shilling attacks

Shilling attacks in recommendation systems involve injecting fake or malicious user profiles into the system to manipulate the generated recommendations [14, 15]. By doing so, the attacker can achieve their goals by benefiting from the attack or impacting the reputation of a business. This section highlights the most commonly used attack models, as well as the role of generative learning in generating realistic ratings

A *Shilling attacks*

Shilling attacks, in general, can be classified based on their intent, i.e. push or nuke attacks [16]. Push attacks tend to increase the ranking of a target item, while nuke attacks aim to demote a specific target item. To achieve a successful attack, attackers mimic the behaviours of genuine users by designing profiles that become the nearest neighbours to real users, as in a collaborative filtering recommender system, to the real users [17]. An attack profile contains four categories of items: selected items, filler items, unrated items, and a target item [18].

Shilling attacks require an attack budget, which is determined primarily by two factors: the size of the attack (the number of injected fake profiles) and the size of each profile (the number of non-zero ratings per profile) [19]. Larger attacks tend to be more effective, but they also increase the cost and risk of detection.

The attacker's level of knowledge about the system significantly influences the design and effectiveness of shilling attacks [20]. This knowledge can be categorized into two main types:

- **User-item interaction knowledge** refers to the knowledge of the dataset used in the training phase of a recommender system. Malicious users can have full or partial knowledge of this data, as user-item ratings are often publicly available and accessible.
- **A target recommender system** refers to the system used to generate recommendations. Some work in the literature assumes complete knowledge of the used recommender system model, while others assume no specific knowledge of the model. Normally, the used models are not accessible to everyone, which gives attackers no specific information regarding the methods, parameters, etc. used.

Simple heuristic-based attacks are commonly employed in the literature and can effectively target traditional collaborative filtering recommenders [21]. These attacks are called standard attacks, and they can be classified as follows [22]:

- **Random attack**, where in this attack, the selected items (I_s), are kept empty, while the ratings for the filler items (I_f), are randomly chosen with a normal distribution centered around the mean. The random attack is considered simple; however, it is not effective.
- **Average attack** is similar to the random attack, except that the ratings for the filler items (I_f) are centered around each item's specific average ratings. To perform the average attack, knowledge of the mean and standard deviation of the items' ratings is required.
- **Bandwagon attack**, in which the selected items (I_s) include popular highly rated items. These attacks aim to mimic the profiles of genuine users who rate popular items.
- **Segment attack**, which sets the selected items (I_s) to be similar to the target item (I_t). When used to push an item; high ratings are assigned to the selected items to increase their similarity to the target items, while filler items are rated low.

These standard attacks require injecting a large number of profiles in the hope of achieving a successful one, for example, 10% of the data in the system, making the attack easily detectable by simple shilling attack detectors [23]. Moreover, fake profiles follow a rating distribution, which differs significantly from the real ones, making the attacks easier to detect. Given the limitations of heuristic-based attacks, particularly their high detectability and limited realism, researchers have begun exploring generative models, such as Generative Adversarial Networks (GANs), to synthesize fake profiles that closely resemble genuine user behavior. Table 1 shows the execution details of these commonly used attacks.

B *Shilling attacks impact*

Shilling attacks significantly threaten the integrity of recommendation systems, impacting users, companies, and the overall system's performance. The impact can be classified based on three perspectives (i) User-Trust, (ii) Business reputation, and (iii) Recommender system's integrity. Our proposed model aims to strengthen existing shilling attack detectors by encouraging the development of more effective ones that can easily detect obfuscated attacks.

- **User-trust**: Shilling attacks can severely impact user trust in recommendation systems. When users receive recommendations that have been manipulated, their satisfaction with the system or company declines, which can lead to reduced engagement and abandonment of the platform that offers the products [24].
- **Business reputation and revenue**: Shilling attacks that manipulate systems can lead to a loss of reputation, directly impacting customer loyalty and potential revenue. Companies often invest in ensuring that their systems are robust and deliver accurate recommendations [25].
- **Recommender system's integrity**: Shilling attacks can degrade the performance of recommendation systems by introducing biases to the data the system is trained

Table 1 Popular standard shilling attacks

Attack	Profile	Execution details
Random	Selected items (I_s)	$\varnothing$
	Filler items (I_f)	$N\left(\bar{r}, \sigma^2\right)$
	Unrated items (I_u)	Left unrated
	Target items (I_t)	r_{min}, or r_{max}
Average	Selected items (I_s)	$\varnothing$
	Filler items (I_f)	$N\left(\bar{r_i}, \sigma_i^2\right)$, item i
	Unrated items (I_u)	Left unrated
	Target items (I_t)	r_{min}, or r_{max}
Bandwagon	Selected items (I_s)	Popular items
	Filler items (I_f)	$N\left(\bar{r_i}, \sigma_i^2\right)$, item i
	Unrated items (I_u)	Left unrated
	Target items (I_t)	r_{max}
Segment	Selected items (I_s)	Items similar to target items
	Filler items (I_f)	Random items, rated with low ratings
	Unrated items (I_u)	Left unrated
	Target items (I_t)	r_{max}

$\boldsymbol{N}$: *Normal distribution*, $\boldsymbol{\sigma}^2$: *variance*, $\bar{\boldsymbol{r}}$: *average rating* on. This manipulation can skew the recommender's performance, leading to less accurate recommendations for all users in the system [26].

3 Related work

Shilling attacks have been widely researched over the past decade, covering various generations and detection methods [27]. This section discusses related work on the detection of shilling attacks in the literature, as well as the role of generative models in enhancing the generation of more effective shilling attacks.

A. *Shilling attacks detection*

Generally, the detection of shilling attacks is classified into two main categories, i.e., supervised and unsupervised techniques. Supervised methods rely on a large set of labeled data to train the classifier, which is used to identify the attack profiles. Meanwhile, unsupervised methods are used to analyze similarity patterns in attack profiles. For instance, in [28], the authors propose a novel approach using bisecting K-means clustering, which leverages the time concentration of group shilling behaviors, where the attackers collaborate within a specific time frame to manipulate the recommendations. The introduced method analyzes item ratings within specific time periods, to improve the chances of detecting similar behaviors. The model is validated by conducting experiments on the Netflix and Amazon datasets. In addition, Zhou et al. [29] address the vulnerability of collaborative recommender systems to shilling attacks by proposing a two-phased detection method called SVM-TIA, which combines support vector machines (SVM and group characteristics analysis. In the first phase, the researchers apply the borderline-SMOTE technique to mitigate class imbalance in the training data, while the second phase fine-tunes the detection by analyzing target items within the detected attack profiles. This approach was tested on the MovieLens 100 K dataset. In [30], the authors introduce a novel approach for detecting shilling attacks in socially-aware networks (SAN), called CNN-SAD. This convolutional neural network-based method extracts detailed features from user-rating profiles. The model provides precise and efficient detection of shilling attacks, and the authors claim that it is also highly effective in detecting obfuscated attacks. In [31], the shilling attacks problem is addressed by an enhanced method that combines a modified Support Vector Machine (SVM), with a Gaussian Mixture Model (GMM). The GMM improves the system's detection rate and reduces dimensionality. Some detection mechanisms generally rely on identifying genuine user profiles based on user traits. In contrast, the detectors primarily examine user behaviour for any irregularities that may suggest the profile is fake. The user's behaviour can be measured by similarity scores, where fake profiles exhibit unrealistic similarity to a significant portion of their neighbouring profiles [32].

B. *Shilling attacks, the generation*

As shilling attacks pose a significant threat to the integrity and reliability of recommender systems, it is crucial to investigate the methods that generate them, particularly those that mimic genuine users' behaviour, to develop more robust defence mechanisms. For instance, the authors in [9] address the limitations of the existing shilling attack detection methods in collaborative filtering-based recommenders, which are unable to identify unknown attacks. They present a novel attack model called the Obscure attack, designed with features that allow it to bypass traditional detection techniques. The model proposed by the researchers is claimed to be highly effective in manipulating the top-N recommendations, by successfully promoting target items while removing genuine ones from the list. Another work is presented in [33], focusing on the segment attack, a type of push attack that manipulates recommendations by targeting a specific set of items. The authors generate segment attacks to evaluate the robustness of several collaborative filtering techniques, including binary, hybrid, user-based, and item-based collaborative filtering recommenders. The experiments were conducted using the MovieLens 100 K dataset, while the robustness was measured by prediction shift and accuracy. A robust collaborative filtering-based recommender system is developed in [34], where the researchers recalibrate user similarity over different periods, ensuring that the selected neighbours reflect the target user's current preferences, which can overcome standard shilling attacks and manipulation. The effectiveness of the proposed algorithm is evaluated on the MovieLens dataset.

C. *Generative models*

Generally, the machine learning models can be classified into discriminative and generative [35]. Discriminative models learn to predict output based on labeled training instances. In contrast, generative models learn to join the training data distribution, allowing them to simulate how data is generated in the real world. Many generative models, such as GANs and Variational Autoencoders (VAEs), are used [36, 37]. This subsection focuses on GANs, which serve as the core of our proposed method. Generative Adversarial Networks (GANs) are a powerful class of generative models that involve two neural networks: the generator (G) and the discriminator (D), which engage in a dynamic adversarial training process [38]. The discriminator (D) is trained to assign a value of 1 to the real data and otherwise, while the generator (G) is trained to produce data with a value of 1 from the classification results. This process follows an objective function, known as the minmax game, presented in Eq. 1. By maximizing the function for the discriminator and minimizing it for the generator, the desired outcome is achieved.

$$\min_{\theta_G} \max_{\theta_D} \mathrm{E}_{\mathrm{x} \sim p_{\mathrm{data}}(\mathrm{x})} \left[\log \mathrm{D}(\mathrm{x})\right] + E_{z \sim N(0,I)} \left[\log\left(1 - D\left(G\left(z\right)\right)\right)\right] \tag{1}$$

where:

- $\mathrm{z} \sim N(0, I)$: Random noise vector.
- $E_{x \sim p_{data}(x)}$: Expectation over the real data samples.

GAN could overcome some of the limitations of previous generative models [39] and was adopted in the literature to generate realistic-looking shilling attacks. For instance, in [40], the authors introduce an Augmented shilling attack framework named AUSH, which uses GANs to create sophisticated and tailored shilling attacks. The proposed model enables attackers to design targeted attacks based on specific goals and intentions. An extended work was introduced in [41] to address the poor attack transferability by learning real user behaviour patterns from sampled data and generating fake profiles with discrete ratings that closely mimic those of genuine users.

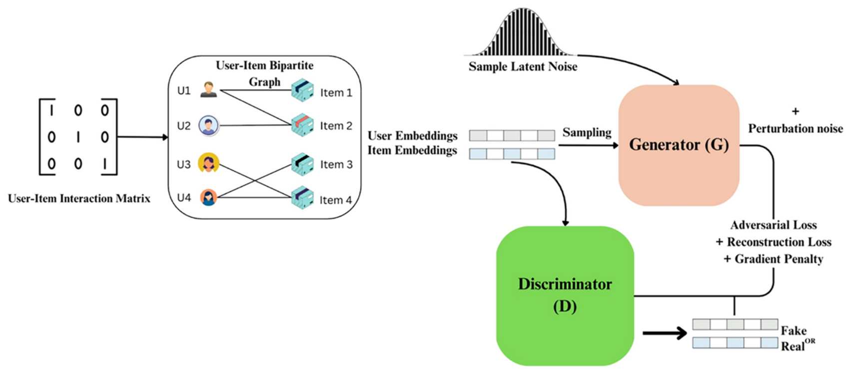

Fig. 2 Overview of the PGAN framework combining GCN embeddings with adversarial generation with latent perturbation

Table 2 Used symbols

Symbol	Description
U	Set of all users
σ	Noise parameter used for generating perturbations
A	Adjacency matrix representing user-item interactions
$H^{(l)}$	Activation from the l-th layer of GCN
$\widetilde{A}$	Normalized adjacency matrix
$W^{(l)}$	Weight matrix for the l-th layer of GCN
$\sigma(.)$	Used Ac
z	Latent noise vector sampled from a normal distribution
R_u^N	Top-N recommendation list for user u
$\nabla_{\widehat{x}} D\left(\widehat{x}\right)$:	Gradient of the discriminator

4 The proposed PGANs attack generation model

Our proposed model, the PGAN, is designed to generate realistic-looking shilling attack profiles, leveraging Generative Adversarial Networks (GANs), with enhanced noise perturbation techniques, as explained in Algorithm 1 and illustrated in Fig. 2. The generated fake profiles aim to bypass traditional detection methods and efficiently bias the recommendation system towards specific attack intents, i.e., pushing or nuking attacks. The PGAN model consists of (i) Bipartite Graph construction for users, items as nodes and edges as ratings; (ii) GCN to extract rich user-item interaction representations; (iii) GANs with penalization techniques are employed to stabilize the traing of the discriminator; (v) Noise perturbations added to the latent space to add variations to the generated profiles, making them more challenging to be detected. Table 2 highlights the main symbols used in the paper.

Input: User-Item interaction matrix, noise parameter σ, number of users *num users*, number of items *num items*
Output: Embeddings, parameter set θ for generator G, parameter set ϕ for discriminator D, Fake profiles
Step 1: Bipartite Graph Construction, and Feature Extraction Using GCNs

1. **Bipartite Graph and GCN initialization:**
   Define two matrices:
   - A: Adjacency matrix representing the user-item interactions.
   - X: Feature matrix where each row represents a user or item.
2. **GCN Layers: for** *each layer in GCN* **do**
   Compute the activation for the next layer as:
   $H^{(l+1)} = \sigma\left(\tilde{A}H^{(l)}W^{(l)}\right)$
   Where:
   $\tilde{A}$ is the normalized adjacency matrix.
   $H^{(l)}$ is the activation from the previous layer.
   $W^{(l)}$ is the weight matrix.
3. **Output:**
   Extract the item and users feature embeddings from the GCN

Step 2: GAN-Based Attack Generation

1. **Initialize GAN:**
   Define Generator (G) and Discriminator (D)
   - G takes latent noise and outputs fake user profiles.
   - D Classifiers profiles as real or fake.
2. **Training, for** each epoch **do**
   a. Sample latent noise:
      $z \sim N(0,1)$
   b. Generate fake profiles:
      $x_{\text{fake}} = G(z)$
   c. Randomly select real profiles x_{real} from the user-item matrix.
   d. Train Discriminator:
      i. compute adversarial loss: $L_{\text{adversarial}} = E[D(x_{\text{fake}})] - E[D(x_{\text{real}})]$
      ii. Apply **gradient penalty** L_{GP}
      iii. Update Discriminator with total loss: $L_{\text{adversarial}} + L_{GP}$
   e. Train Generator:
      i. Generator total loss: $L_G = -E[D(x_{\text{fake}})] + L_{reconstruction}$

Step 3: Add Perturbations with Noise
For Each Generated profile:
- Add **Gaussian Noise** to introduce perturbations:
  $x_{\text{perturbed}} = x_{\text{fake}} + \mathcal{N}(0, \sigma^2)$

Algorithm 1 PGAN with latent perturbation

A. *Bipartite graph construction*

Bipartite graphs $G = (U, I, E)$ are a type of graph where the vertices are divided into two distinct groups (U, I). Edges E in these graphs only connect vertices from one group to vertices in the other one. In our model, one of these two groups represents users, while the other represents items, with edges assigned to represent the interactions, i.e., the ratings that users have given to items [42]. While the traditional matrix-based methods in recommendation systems, such as collaborative filtering methods, mainly focus on direct user-item interactions, the bipartite graph-based method, incorporating Graph Convolutional networks (GCNs), can leverage higher-order interactions. The GCNs propagate information from a specific user to an item through direct interaction and between items that are indirectly connected through similar users, thereby capturing more complex patterns in user preferences can be captured [43]. Bipartite graphs are represented by an adjacency matrix A, structured as:

$$A = \begin{bmatrix} 0 & B \\ B^T & 0 \end{bmatrix} \tag{2}$$

where B is an $|U| \times |I|$ Matrix. Each entry B_{ui}, in this matrix, denotes the presence and strength of the interaction between user u and item i, such as a rating or a review. The zero blocks in A indicate no direct interactions between users or items, aligning with the bipartite nature. By transforming the adjacency matrix, GCNs can perform convolution operations on local neighborhood data.

B. *Graph convolutional networks (GCN) for user-item embedding*

Graph Convolutional Networks (GCNs) enhance the semantic understanding of user-item interactions within our model. By treating these interactions as a bipartite graph, where nodes represent users and items, and edges denote the interactions, such as the ratings, GCNs enable a more in-depth analysis than traditional methods [44]. Traditional collaborative filtering often struggles with capturing complex and higher-order relationships effectively; however, the GCNs perform well in this area by utilizing the graph structure to propagate information across all connections in the data, improving the recommendation accuracy by capturing both immediate and neighboring interactions [45]. Each layer of the GCN aggregates information from a node's neighbours, which helps lead to more meaningful embeddings for each node, and then embeds both user and item feature representations. These embeddings are capable of reflecting users preferences and items' attributes. GCN can integrate local neighborhood data with overall graph dynamics, thus enhancing the model's ability to generalize beyond the observed interactions. The user and item embeddings obtained from the final GCN layer are concatenated and used as input features for the PGAN generator, allowing it to synthesize fake profiles based on realistic interaction structures. The inclusion of GCN enables the model to leverage higher-order connectivity patterns among users and items, which enriches the representation space. WGAN-GP further ensures stable adversarial training and mitigates gradient vanishing, resulting in improved convergence and increased attack realism, as explained in the subsequent subsections.

C. *PGAN architecture and training*

Although Generative Adversarial Networks (GANs) have been considered powerful tools for generating realistic synthetic data, they face instability during training, leading to vanishing gradients [46], a problem where the gradient used in updating the network weights during backpropagation tends toward zero, halting the learning process. To address this issue, our model introduces a penalty term to the discriminator's loss function: Wasserstein GAN with Gradient Penalty (WGAN-GP) [47]. In the proposed model, the generator (g) and discriminator (D) networks are designed to play an adversarial game, where the generator produces fake profiles, and the discriminator tries to differentiate between real and fake profiles. The generator (G) takes the latent vector $z$sampled from a standard normal distribution and produces an output corresponding to a fake user profile, where each value represents a rating. The generator aims to learn the underlying distribution of genuine profiles such that the generated profiles are indistinguishable from the real ones. On the other hand, the discriminator (D) receives real or fake profiles as input, where it evaluates them and outputs a probability score that classifies the inputs as real or fake. This setup enables the discriminator to learn the distinguishing features between authentic and synthetic data, pushing the generator toward producing more realistic profiles. Adding the gradient penalty to the WGAN-GP framework helps prevent the discriminator from converging too quickly, thereby balancing the training process. It mitigates the risk of the generator gradients vanishing problem.

D. *Loss functions*

To ensure effective and robust training, we employ multiple loss functions that work in tandem, including adversarial loss, reconstruction loss, and gradient penalty, to facilitate stable training.

- **Adversarial loss (Wasserstein loss)** measures the discrepancy between the distributions of genuine and generated profiles [48]. Minimizing the Wasserstein distance makes the generated distribution as close to the real ones [49]. This also helps to stabilize the training process and improve the quality of the generated data.

$$L_{\text{adversarial}} = E\left[D\left(x_{\text{fake}}\right)\right] - E\left[D\left(x_{\text{real}}\right)\right] \tag{3}$$

where,

- $D\left(x\right)$: Discriminator output.
- $x_{\text{fake}}, x_{\text{real}}$: Fake and real profiles.
- **Reconstruction loss** ensures that the generated profiles resemble the real ones in distribution, indicating that the generator could capture the patterns of the actual user profiles [50]. In our proposed model, the Mean Squared Error (MSE) loss is employed as the reconstruction loss, as explained in Eq. 4. The MSE loss facilitates the model's training by penalizing the discrepancies between the generated and real profiles. The sum of the squared difference is normalized by the total number of profiles N.

$$L_{\text{reconstruction}} = \frac{1}{N}\sum\nolimits_{i=1}^{N}\left(x_{\text{fake}}^{i} - x_{\text{real}}^{i}\right)^{2} \tag{4}$$

where,

- N: The number of user profiles.
- $x_{\text{fake}}^{i}, x_{\text{real}}^{i}$: Fake and Real profiles, respectively.
- **The gradient penalty (WGAN-GP)** is introduced here to avoid the vanishing gradient problem during training, which helps stabilize the training process and prevent overfitting by maintaining the discriminator gradient within a reasonable range.

$$L_{\text{GP}} = \text{E}\left[\left(\left|\nabla_{\widehat{x}}D\left(\widehat{x}\right)\right|_{2} - 1\right)^{2}\right] \tag{5}$$

where,

- $\widehat{x}$: The interpolated sample between genuine and fake profiles.
- $\nabla_{\widehat{x}}D\left(\widehat{x}\right)$: the gradient of the discriminator's output with respect to the interpolated sample.
- **Total learning loss**, is the combination of the adversarial loss, reconstruction loss and gradient penalty.

$$L_{\text{Total}} = L_{\text{adversarial}} + L_{\text{reconstruction}} + L_{\text{GP}} \tag{6}$$

E. *Noise perturbation technique in latent space*

To generate more realistic-looking and undetectable fake profiles, a Gaussian noise perturbation is applied to the generator's output latent space. Noise perturbation introduces randomness and variability into the generated fake profiles to mimic the real users behaviors. For instance, in real-world scenarios, the users' preferences and interactions fluctuate over time, and by adding noise, the generated profiles can exhibit similar variations, making them look more authentic. Equation 7 explains the added perturbation noise for the generated fake profiles.

$$z_{\text{perturbed}} = z + \epsilon \cdot \mathcal{N}\left(0, \sigma^{2}\right) \tag{7}$$

where,

- z: Latent Space Vector (i.e. generated user profiles representation).
- ϵ: Perturbation intensity.
- $\mathcal{N}\left(0, \sigma^{2}\right)$: Gaussian noise with zero mean.

F. *Surrogate recommender system*

A surrogate recommender model is employed to evaluate the effectiveness of our proposed shilling attack approach. This model enables us to simulate the impact of the attack. The regularized Matrix Factorization model is used, due to its common adoption [51].

G. *Evaluation framework*

To evaluate our PGAN model, we employ a range of metrics to assess its effectiveness. These metrics allow us to measure the impact of the generated fake profiles on the recommendation system. The model is also compared against the state-of-art models. In addition to that, we compare the attack effectiveness against common traditional shilling attacks, such as (random, average, bandwagon and segment) attacks.

- **Hit rate (HR)** [52] measures the number of users that receive the target item in their top-N recommendation list after the attack, as calculated in Eq. 8, where U is the set of users, and i is the target item. R_u^N is the top-N recommendation for a user.

$$HR = \frac{1}{|U|}\sum_{u \in U} 1\left(i \in R_u^N\right) \tag{8}$$

- **Prediction shift** [53] quantifies the difference between the predicted ratings for a target item before and after an attack.
- **F-measure**, which measures how well an attack misleads the detection mechanisms. The harmonic mean between precision and recall is calculated as shown in Eq. 9.

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{9}$$

5 Experimental results

This section presents the experimental setup and results, which are used to evaluate the proposed model. The model is evaluated against the state-of-the-art methods presented in these papers [40] (AUSH) and [41] (Leg-UP). Additionally, the effectiveness of the

Table 3 Used datasets description

Dataset	MovieLens 100 K	Amazon-electronics
Unique Users	943	657
Items	1,682	6016
Total Ratings	100,000	70,652
Ratings Range	1 to 5	1 to 5

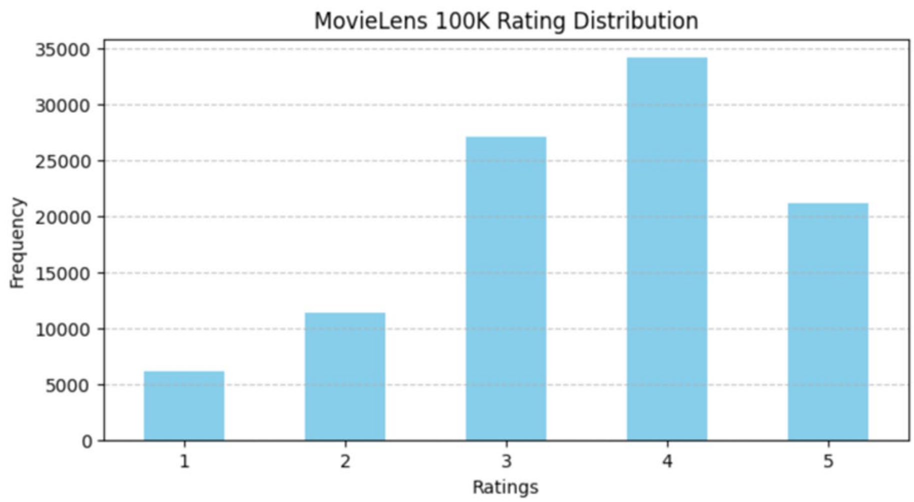

Fig. 3 MovieLens 100 K rating distribution

PGAN-generated fake profiles is evaluated against traditional attacks, namely random, average, bandwagon, and segment.

A. *Datasets*

To evaluate our proposed model, we utilize two datasets: MovieLens100K [12] and Amazon Electronics, a subset of the Amazon dataset [13]. Table 3 highlights the main statistics for both datasets. Figure 3 illustrates the rating distribution for the MovieLens 100 K dataset. The distribution shows a clear preference for higher ratings, with ratings of 4 being the most frequently used value. On the other hand, ratings of 1 and 2 are relatively infrequent, suggesting that users are generally less likely to assign these lower ratings. Additionally, Fig. 4 illustrates the rating distribution for the Amazon electronics dataset, indicating a bias towards high ratings, specifically ratings of 5.

B. *Learning user-item graph embeddings with GCN*

In the PGAN with noise perturbations model; a GCN is trained on MovieLens 100 K and Amazon product reviews to model the user-item interactions. The user-item interaction matrix is used to construct a bipartite graph, where users and items are represented as nodes, and the interactions are represented as weighted edges. Figure 5 depicts a bipartite graph representation of the top 10 user-item interactions within the MovieLens dataset. Each edge represents the rating given by users for the corresponding items. Figures 5 and 6 highlight how certain users and items are more heavily interconnected, suggesting patterns crucial for training the PGAN to generate realistic-looking synthetic interactions; for instance, darker edges indicate higher interaction weight.

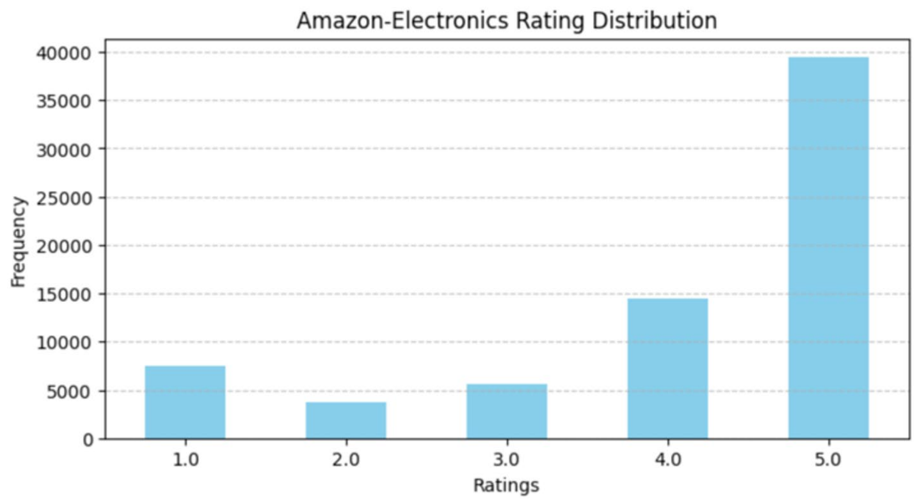

Fig. 4 Amazon dataset rating distribution

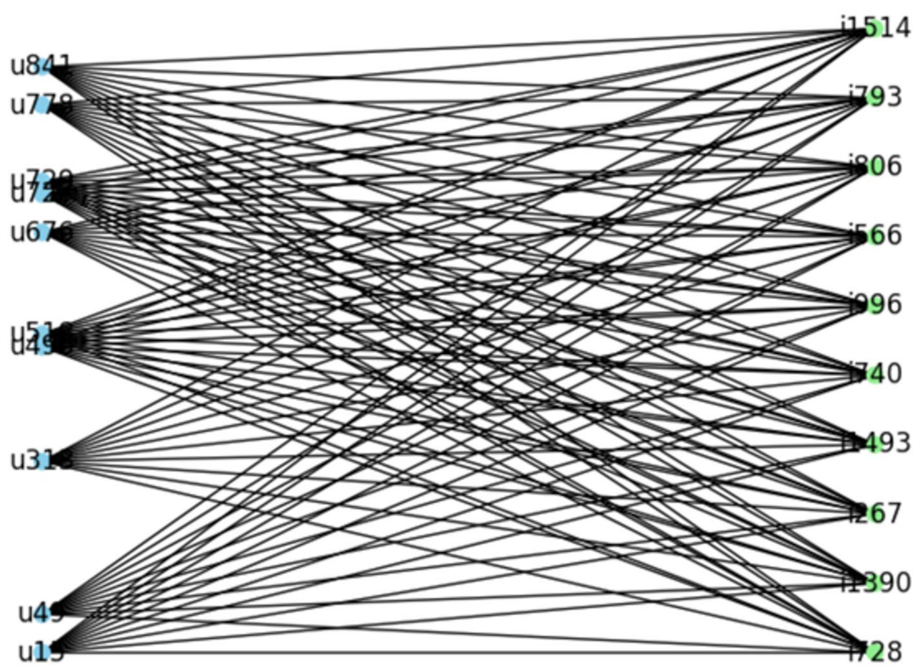

Fig. 5 Bipartite graph representation for the Top 10 user-item interactions (MovieLens)

The interaction matrix is then transformed into an adjacency matrix, which is fed to the GCN. It is transformed into a normalized adjacency matrix in the graph convolution operation, indicating the connections between users and items, including self-loops. That transformation contains the relationship between nodes, which are then learned through the GCN layers. The used GCN model consists of two convolutional layers, initialized with 32-dimensional embeddings for users and items. Training for 100 epochs, the GCN exhibited a rapid decrease in loss, converging from its initial high values to 0.0068 for MovieLens, as shown in Fig. 7, and to 0.00032 for the Amazon dataset, as illustrated in Fig. 8. The losses during training for both the Amazon and MovieLens datasets demonstrate the model's ability to effectively learn user and item embeddings.

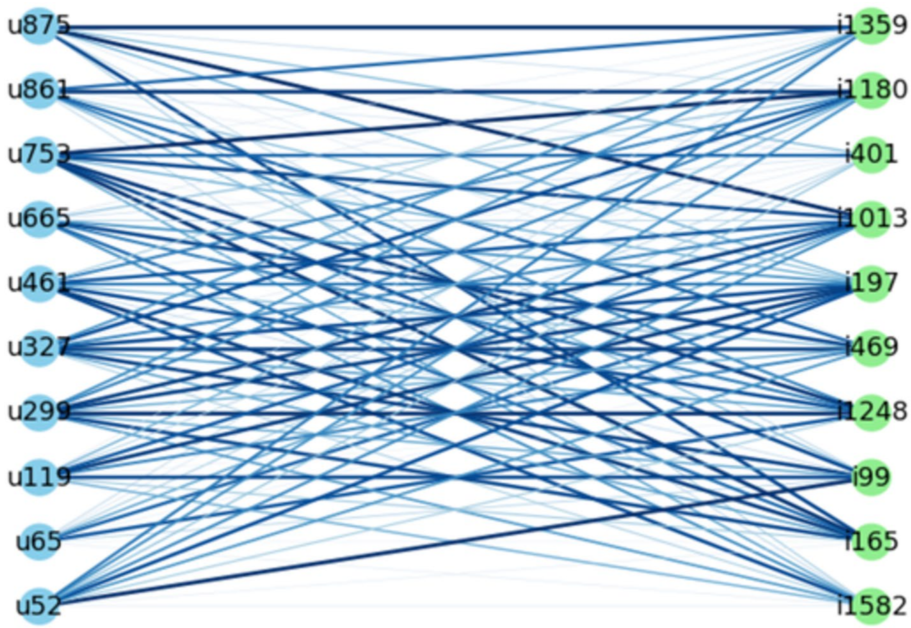

Fig. 6 Bipartite graph representation for the top 10 user-item interactions (Amazon)

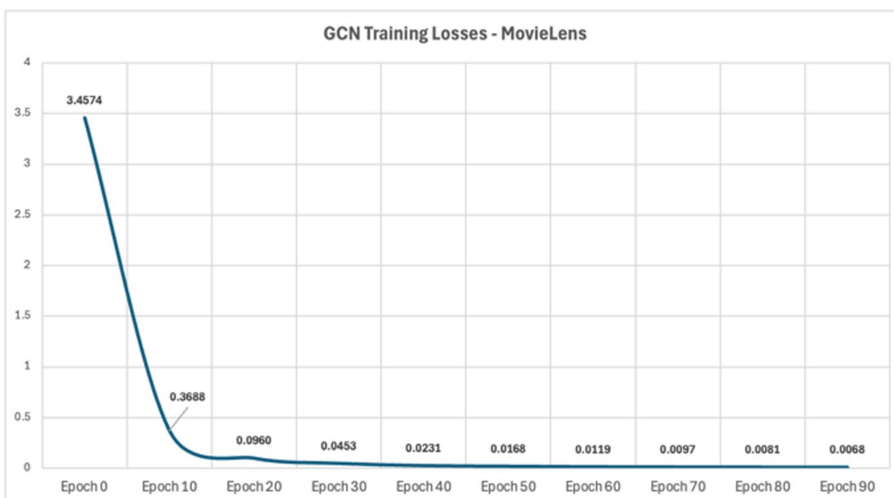

Fig. 7 GCN losses – MovieLens 100 K

C. *GAN training with and without penalization*

In the training phase, we incorporated multiple loss functions to generate realistic user profiles, including adversarial loss, reconstruction loss, and gradient penalty. Figure 9 represents the GAN without penalization; the generator's loss increases over time, while the discriminator's loss drops to near-zero values, which indicates overfitting.

In Fig. 10, including a gradient penalty helps balance the learning process, as presented in the discriminator-balanced training. Figure 11 illustrates the GAN architecture used, displaying the number of layers, nodes, and other relevant details.

D. *Generated fake profiles distribution*

Fig. 8 GCN losses – amazon

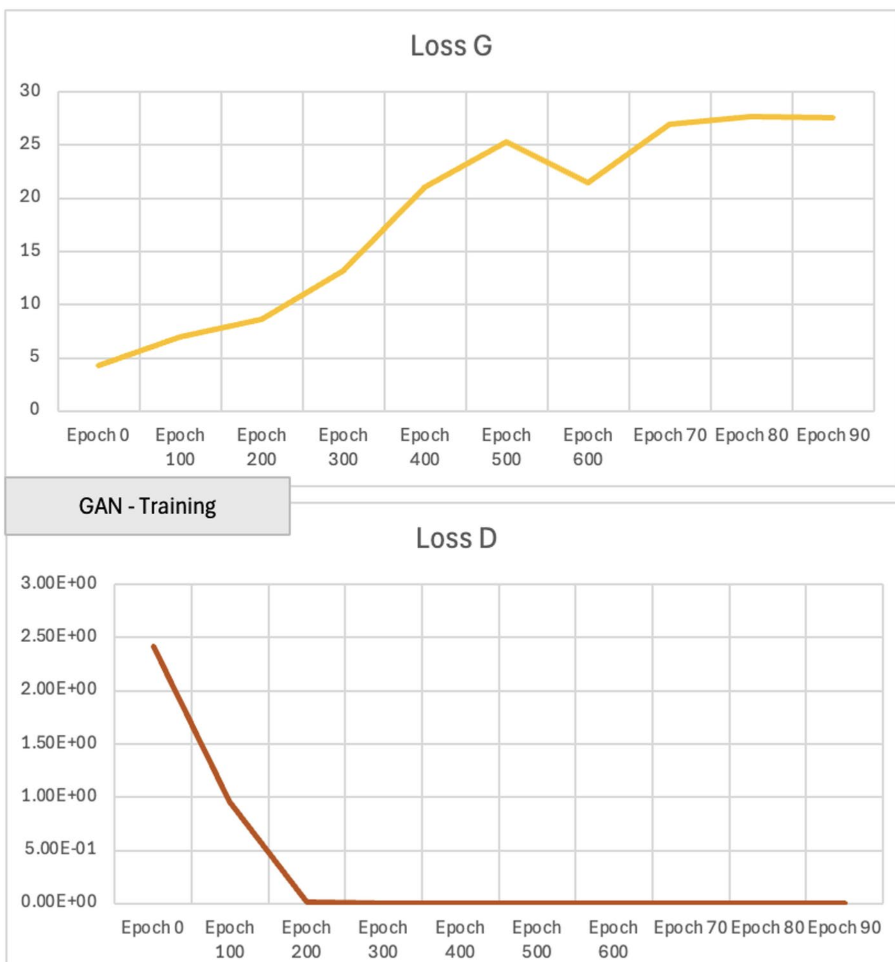

Fig. 9 MovieLens GAN training with no penalization

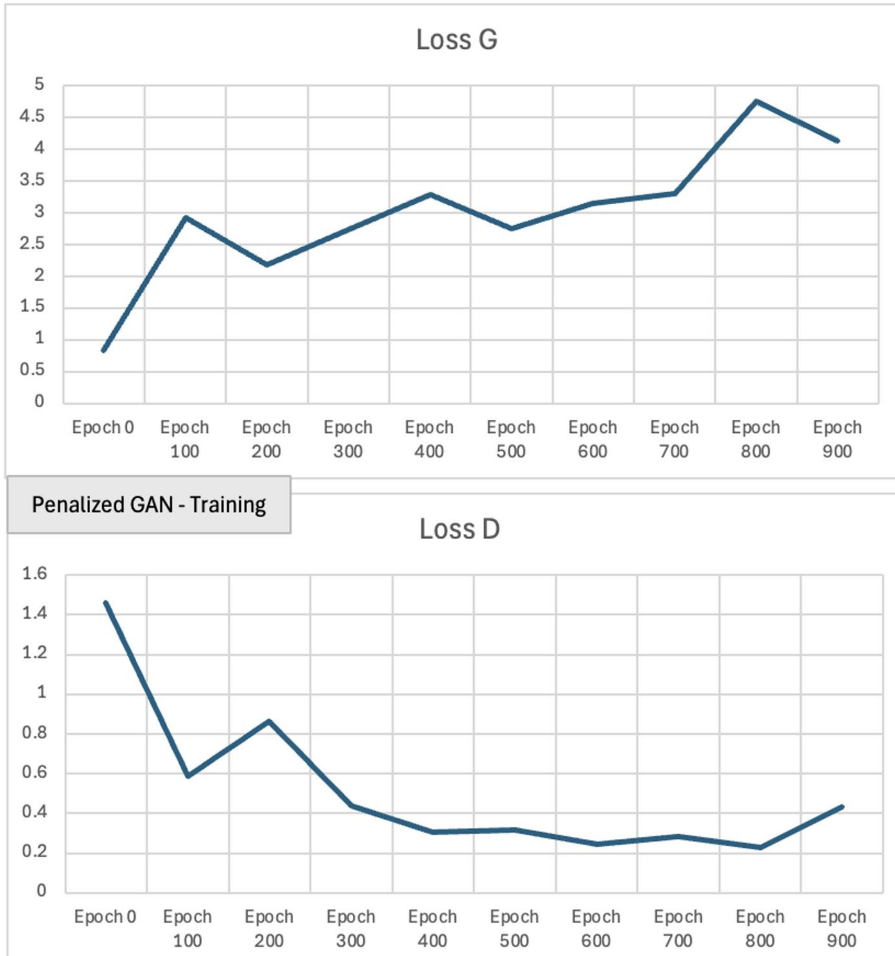

Fig. 10 MovieLens GAN training with gradient penalty

In the PGAN with noise perturbations model, a fake profile distribution analysis is performed for both the MovieLens 100 K and Amazon datasets. The analysis aims to understand how well the fake profiles generated by our PGAN model mimic real user profiles against those generated by traditional heuristic methods, such as Average, Bandwagon, and Segment attacks. As observed in Fig. 12 for the MovieLens dataset and Fig. 13 for the Amazon dataset, the proposed model successfully generates fake profiles that are more dispersed with real ones compared to the traditional attacks. In traditional attack models, such as bandwagon and segment, fake profiles are easily detected by any shilling attack detector. For instance, the Segment attack profiles typically display a clustered distribution, where ratings are maximized for a specific subset of items. This clustering targets the visibility of certain items, thus making the fake profiles easily identifiable. On the other hand, the Bandwagon attack utilizes popular items, assigning them the highest ratings to manipulate the recommendations, resulting in a more concentrated pattern of ratings.

E. *Performance of the PGAN-generated attacks*

The effectiveness of PGAN-generated fake profiles is assessed against several baseline shilling attack techniques, including Bandwagon, Average, Random and Segment

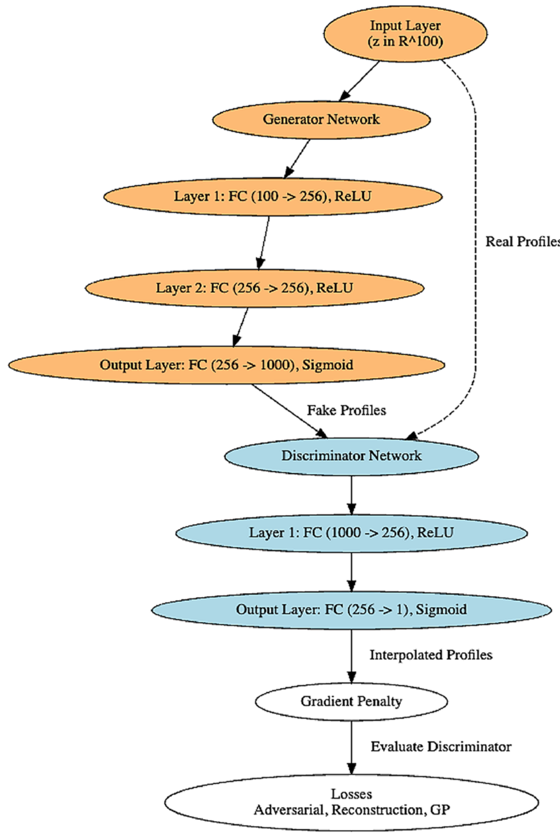

Fig. 11 Detailed adopted GAN architecture. The diagram presents the configuration of the generator and discriminator Networks, showing the dimensions of the input, hidden, and output layers

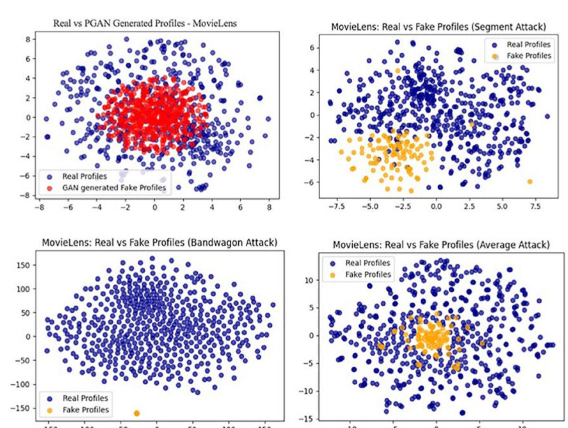

Fig. 12 MovieLens fake and real profiles

attacks. A Regularized Matrix Factorization RS surrogate model is used to evaluate the impact of the attacks. 5% of the user profiles are generated using the PGAN model, and HR@10 is computed to measure the percentage of times a target item appears in the top 10 recommendations. Additionally, the average change in predicted ratings for target items before and after the attack is calculated as the prediction shift. The higher the prediction shift, the stronger the indication that a successful attack has been implemented. As presented in Table 4, The Hit Rate@10 for the PGAN model reaches 0.2051,

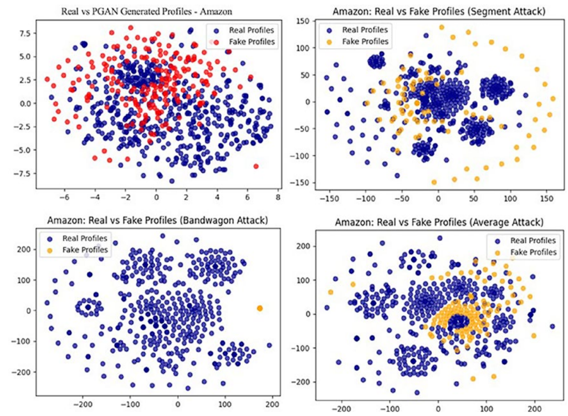

Fig. 13 Amazon fake and real profiles

Table 4 HR@10 and prediction shift benchmark-MovieLens

Metric	Hit rate@10	Prediction shift
Dataset	MovieLens 100 K	
PGAN	**0.2051**	**0.3734**
Bandwagon attack	0.178134	0.0041
Average attack	0.19833	0.00012
Random attack	0.125905	0.00195
Segment	0.138162	0.00132
AUSH [40]	0.1849	N/A
Leg-UP [41]	0.1916	N/A

Table 5 HR@10 and prediction shift benchmark-amazon

Metric	Hit rate@10	Prediction shift
Dataset	Amazon electronics	
PGAN	**0.2076**	**0.5034**
Bandwagon attack	0.1452	0.000867792
Average attack	0.156	0.000064053
Random attack	0.17	−0.0012
Segment	0.13	−0.00147
AUSH	N/A	N/A
Leg-UP	N/A	N/A

surpassing both the baseline and state-of-the-art models' attacks. In terms of prediction shift, PGAN shows a significant shift of 0.3734, indicating that the model can alter the predicted ratings of target items. This shift surpasses the rest of the baseline attacks, which show minimal impact on the recommendation system. Similarly, in the Amazon dataset, presented in Table 5, the PGAN proposed model achieved the highest HR@10 of 0.2076 and a significant Prediction shift of 0.5034, indicating its strong ability to successfully mimic user profiles and influence recommendations. The bandwagon attack performed with a moderate HR@10 of 0.1452, but its low Prediction shift suggests that the attack struggled to effectively manipulate the recommendations. Random and segment attacks on the Amazon dataset have the least impact on the recommendations.

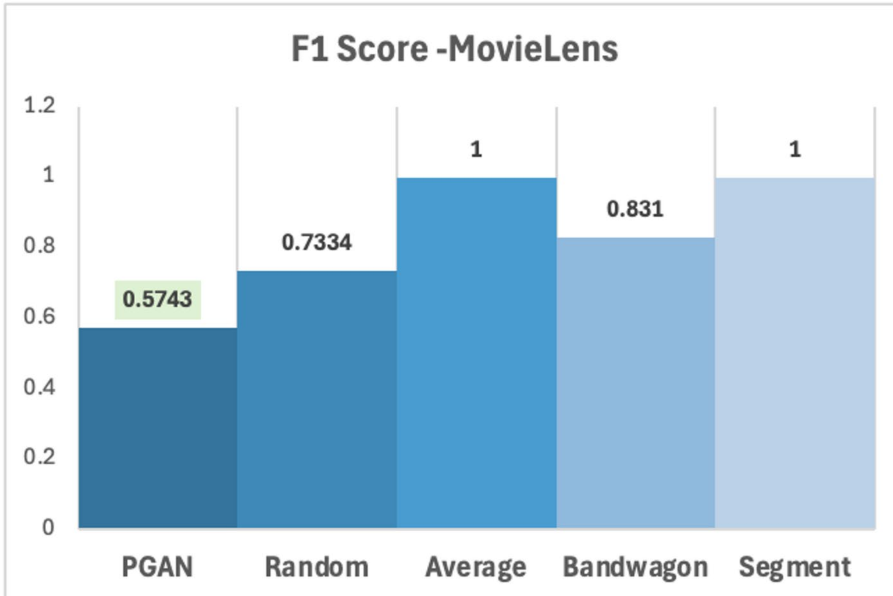

Fig. 14 F1-score benchmark MovieLens

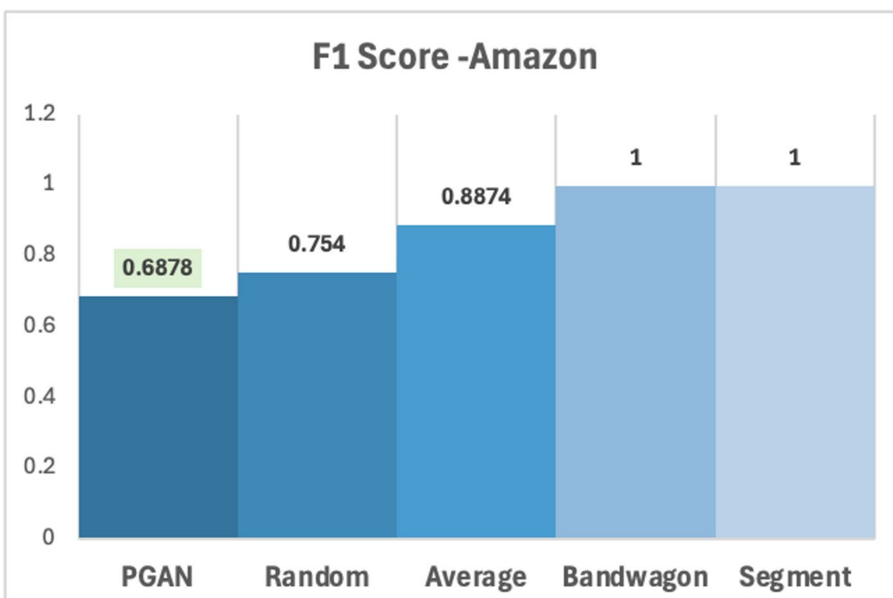

Fig. 15 F1-score benchmark amazon

F. *Attack detection/Invisibility*

To evaluate how well PGAN-generated attacks can mislead classifiers of shilling attacks, we trained a random forest model and calculated the F1-score for detection. The results presented in Figs. 14 and 15 show that the PGAN proposed model achieved the lowest F1 scores, with values of 0.5742 for MovieLens and 0.6878 for Amazon, indicating that the fake profiles generated by our model are more difficult to distinguish from real ones. On the other hand, baseline attacks, such as random, average, and segment, had much

Table 6 Comparison of stabilization techniques-MovieLens

Variant	Mean D loss	D loss variance	Mean G loss
Baseline GAN	1.3151	0.0006	0.7051
GAN + Gradient clipping	1.281	0.0004	0.75
GAN + LeakyReLU	1.3426	0.0005	0.6933
WGAN-GP	**2.3025**	**1.6183**	**0.603**

higher F1 scores, which could reach 100%, indicating that a simple classifier can easily distinguish these heuristic attacks.

G. *Ablation study*

To validate the choice of WGAN-GP as the stabilization method in the proposed PGAN framework, we conducted an ablation study comparing WGAN-GP to alternative stabilization techniques commonly used in GAN training, including gradient clipping, LeakyReLU activation, and the baseline GAN configuration without any stabilization. All experiments were conducted on the MovieLens dataset for consistency. Each variant was trained for 10 epochs using the same architecture and training parameters, with the only difference being the stabilization strategy applied to the discriminator. The evaluation focused on the discriminator loss mean and variance across epochs, as well as the generator loss. Higher discriminator loss variance reflects more active adversarial dynamics, which are critical to effective profile generation. As demonstrated in Table 6, WGAN-GP achieves higher discriminator loss variance and a lower generator loss, consistent with more sustained adversarial learning.

6 Discussion

In the proposed PGANs, the GCN and GAN with noise perturbations generate realistic-looking ratings in the recommendation system. The proposed model demonstrated a high capability to mimic real user behavior compared to traditional attack methods such as bandwagon, average, and segment attacks. By integrating noise perturbations, the generated profiles exhibited greater diversity, resembling the complexity of genuine ones. The undetectability of our model's attack is also tested by examining the distribution of the generated profiles, where the dispersed distribution complicates the task of detecting shilling attacks. Incorporating GCN embeddings and WGAN-GP stabilization increases the computational requirements relative to a standard GAN. The GCN passes scale, and the gradient penalty requires additional backward computations per batch. In practice, training PGAN on the MovieLens dataset required approximately 5 min per epoch, compared to 2 min for the baseline GAN, reflecting a manageable overhead. Furthermore, the proposed model outperforms state-of-the-art methods in the MovieLens and Amazon datasets, indicating that the PGAN with noise perturbation model can be generalized to other datasets and recommendation systems. However, the transferability of this approach to even more diverse or larger datasets remains an area for further investigation.

7 Conclusion and future directions

In this paper, we address the issue of distinguishable shilling attacks by proposing a novel approach for shilling attack generation, named PGAN, which leverages GCN with GAN and noise perturbations. Our method aims to create realistic-looking fake user profiles that mimic the behavior of real users, which captures the underlying relationships between users and items through the GCN. Additionally, we employ the penalization technique to stabilize GAN training and introduce noise perturbations to enhance the realism of the generated profiles. The proposed PGAN with noise perturbations model could surpass traditional and state-of-the-art attacks. In future work, we aim to test the transferability of this model, by evaluating its ability to adapt to different recommender systems and domains with larger datasets. In addition, exploring different alternatives to noise perturbations, such as adversarial training or dropout perturbations, can increase the indistinguishability and effectiveness of generated profiles.

Author contributions
D N: Conceptualization, Methodology, Coding, Validation, Formal analysis, Investigation, Resources, Data Curation, Writing - Original Draft, Visualization, Supervision.R K: Conceptualization, Methodology, Validation, Formal analysis, Investigation, Resources, DataCuration, Writing - Review & Editing, Visualization, Funding, Project administration.

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This study does not involve any clinical trials; therefore, clinical trial registration is not applicable.

Data availability
The datasets analysed during the current study are available in the public domain: the MovieLens 100 K dataset (https://grouplens.org/datasets/movielens/100k/) and the Amazon Electronics dataset (https://www.kaggle.com/datasets/yasserh/amazon-product-reviews-dataset).

Declarations

Ethics approval and consent to participate
Not applicable.

Consent to publication
Not applicable.

Competing interests
The authors declare no competing interests.

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